pharmaniaga®

Fentanyl 50mcg/mL Injection

COMPOSITION
Each ampoule of 2 mL contains Fentanyl Citrate 157.10 mcg equivalent to Fentanyl 100 mcg.

PRODUCT DESCRIPTION
Clear, colourless solution. The compatible infusion solutions for intravenous use are as follows:
- Sodium Chloride 0.9%solution
- Glucose 5% solution

The solution should be clear, colourless solution when diluted. If not required immediately, the diluted solution may be stored up to 24hours at temperature below $30^{o}C$ and at refrigerated condition ($2^{o}C$ to $8^{o}C$) after mixing with the compatible solutions.

PHARMACODYNAMICS
Fentanyl is a potent opioid analgesic. It acts mainly along with μ-opioid receptor. Fentanyl is used as analgesic supplement to general anaesthesia or as single anaesthetic. It has strong depressive effect on respiration and low cardio-circulatory effect on increasing dose. 100mcg (2.0mL) of dose shows analgesic property equivalent to 10mg of morphine. Fentanyl has rapid onset of action and it takes approximately 30 minutes to exhibit analgesic effect after single dose IV dose. Level of analgesia is dose related and it is altered according to pain level of surgical procedure. Fentanyl can cause muscle rigidity, euphoria, miosis and bradycardia depending upon dose and speed of administration. Fentanyl does not suppress stress induced hormonal change. Histamine assays and skin-wheal testing showed that clinically significant histamine release after fentanyl application is rarely observed. All actions of fentanyl are reversed by opioid antagonist such as naloxone.

PHARMACOKINETICS
Distribution
Fentanyl is a lipid-soluble drug and described as three-compartment model.
Fentanyl has short distribution phase. Rapid onset of action is achieved at high concentrations in well perfused tissues like kidneys, brain and lungs. Sequential distribution half-lives of fentanyl is 1 minute and 18 minutes. It has V_c(volume of distribution of central compartment) of 13L and V_{dss}(distribution volume at steady-state) of 339L. The drug redistributed to other tissues and assemble in skeletal muscle and fat prior to gradual release into blood. Fentanyl has plasma protein binding of 80-85%. Half-life is prolonged in elderly patients or after repeated administration.

Metabolism
Fentanyl metabolised rapidly in liver by N-dealkylation and through cytochrome P450 isoenzyme CYP3A4. The main metabolite is norfentanyl. Fentanyl clearance is 0.5L/hr/kg.

Excretion
The terminal half-life of fentanyl is 3.7 hours. 75% of administered fentanyl dose is excreted through urine within 24 hours. 10% excreted as unchanged drug in the urine.

INDICATION
Fentanyl is a short acting opioid used
- as an intravenous analgesic agent in surgical procedures.
- as an adjunct in the maintenance of generalanaesthesia.
- in combination with a neuroleptic agent in the technique of neuroleptanalgesia.
- as a respiratory depressant/analgesic in patients requiring prolonged assisted ventilation.
- for induction of anaesthesia.

RECOMMENDED DOSAGE
Fentanyl should be given only in an environment where the airway can be controlled and by personnel who can control the airway.
The dose of fentanyl is adjusted individually according to age, body weight, physical status, pathological condition, co-medication and as well as type of surgical procedure and type of anaesthesia.For guidance, the following dosage schedules are proposed. For other special dosage recommendations please refer to the literature.

Neurolept analgesia and neurolept anaesthesia
For neurolept analgesia adults normally will require an initial dose of 50 to 100 micrograms (0.7 - 1.4 microgram/kg) fentanyl, slowly injected intravenously in combination with a neuroleptic (preferably droperidol). If necessary a second dosage of 50 to 100 micrograms (0.7 - 1.4 microgram/kg) fentanyl can be given 30 to 45 minutes after the initial dose.
For neurolept anaesthesia under the condition of assisted ventilation adults in general will require an initial dose of 200 to 600 micrograms (2.8 - 8.4 micrograms/kg) fentanyl slowly injected intravenously in combination with a neuroleptic (preferably droperidol).
The dosage depends on the duration and the severity of the surgical procedure and on the medication used for general anaesthesia. For maintenance of anaesthesia, additional doses of 50 to 100 micrograms (0.7 - 1.4 microgram/kg) fentanyl can be given every 30 to 45 minutes. The time intervals and doses of these additional administrations have to be adjusted according to the course of the surgical procedure.

Analgesic component in general anaesthesia
Adults:
Premedication: 50 to 100 micrograms (1 to 2 mL) fentanyl administered intramuscularly 30 to 60 minutes prior to surgery (modified for high risk patients).
For induction: If fentanyl is used as an analgesic component in general anaesthesia with intubation and ventilation of the patient, in adults initial fentanyl doses of 70 - 600 microgram (1 - 8.4 microgram/kg) can be applied as adjunct to general anaesthesia.
For maintenance of analgesia during general anaesthesia, additional doses of 25 - 100 microgram (0.35 - 1.4 microgram/kg) fentanyl are to be injected subsequently. The time intervals and the dosage are to be adjusted according to the course of the surgical procedure.

Pain management in the intensive care unit
For use in pain management of ventilated patients in the intensive care unit, the dosage of fentanyl has to be adjusted individually, depending on the course of pain and on concomitant medication. Normally the initial doses are in the range of 50 to 100 micrograms i.v. (0.7 - 1.4 microgram/kg) but can be titrated higher if necessary.
The initial dose normally is followed by repeated injections, of up to a total of 25 to 125 micrograms fentanyl per hour (0.35 - 1.8 microgram/kg/h).

Dosage in Paediatric population
Children aged 12 to 17 years should follow the recommended adult dose schedule.For children aged 2 to 11 years the usual dosage regimen is recommended as follows:

	Age in Years	Initial dose in microgram s/kg body weight	Supplement al dose in micrograms/ kg body weight
Spontaneous Respiration	2-11	1-3	1-1.25
Assisted Ventilation	2-11	1-3	1-1.25

Use in children:
Analgesia during operation, enhancement of anaesthesia with spontaneous respiration:
Techniques that involve analgesia in a spontaneous breathing child should only be used as part of an anaesthetic technique, or given as part of a sedation/analgesia technique with experienced personnel in an environment that can manage sudden chest wall rigidity requiring intubation, or apnoea requiring airway support.

Dosage in elderly and weak patients
The initial dose in elderly and weak patients should be reduced. The effect of the initial dose should be taken into account for the determination of supplemental doses.

Dosage in patients with chronic opioid medication
In patients with chronic opioid medication or with a known history of opioid abuse, a higher dosage of fentanyl may be necessary.

Dosage in patients with additional diseases
In patients with one of the following diseases the intended dosage of fentanyl should be titrated very carefully:
- uncompensated hypothyreosis
- lung diseases, especially those with reduced vital capacity
- alcohol abuse
- impaired hepatic function
- impaired renal function

Caution is also required if fentanyl is to be administered to patients with adrenal insufficiency, prostatic hypertrophy, porphyria and bradyarrhythmia.

In all these conditions, except alcohol abuse, the dose may have to be reduced. In alcohol abuse, the dose may have to be either reduced or increased.

In these patients a prolonged postoperative monitoring period is recommended.

ROUTE OF ADMINISTRATION
Parenteral
[For IV or IM use only.]

CONTRAINDICATIONS
Fentanyl should not be used in patients with
- known hypersensitivity to fentanyl, other morphino-mimetics or to any of the excipients.
- respiratory depression without artificial ventilation.
- known intolerance to fentanyl or other morphino-mimetics.
- obstructive airway disease.
- in patients after operative interventions in the biliary tract.

WARNINGS AND PRECAUTIONS
Intravenous fentanyl should only be used by a trained anaesthetist in hospitals or other locations with facilities for intubation and assisted ventilation.
The vital functions of the patient have to be monitored routinely. This also applies to the postoperative period.

Respiratory depression
Fentanyl has a strong dose dependent depressing effect on respiration which may be prolonged especially in the elderly. In neonates, respiratory depression is to be expected already after small doses. As with all potent opioids, respiratory depression is dose related and can be reversed by a specific opioid antagonist, but additional doses may be necessary because the respiratory depression may last longer than the duration of action of the opioid antagonist. Profound analgesia is accompanied by marked respiratory depression, which can persist or recur in the

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postoperative period. Therefore, patients should remain under appropriate surveillance. Resuscitation equipment and opioid antagonists should be readily available. Hyperventilation during anesthesia may alter the patient's responses to CO2, thus affecting respiration postoperatively.

Muscle rigidity
Induction of muscle rigidity, which may also involve the thoracic muscles, can occur, but can be avoided by the following measures: slow IV injection (ordinarily sufficient for lower doses), premedication with benzodiazepines and the use of muscle relaxants, controlled ventilation.
Non-epileptic (myo)clonic movements can occur. Muscle rigidity may also lead to respiratory depression.

Cardiac disease
Bradycardia, and possibly cardiac arrest, can occur if the patient has received an insufficient amount of anticholinergic, or when fentanyl is combined with non-vagolytic muscle relaxants. Bradycardia can be treated with atropine. It is imperative to ensure that adequate spontaneous breathing has been established and maintained before dischargefrom the recovery area whenever large doses or infusions of fentanyl
have been administered.
Opioids may induce hypotension, especially in hypovolemic patients. Induction doses should be adapted and administered slowly, in order to prevent cardiovascular depression. Appropriate measures to maintain a stable arterial pressure should be taken.

Special dosing conditions
The use of rapid bolus injections of opioids should be avoided in patients with compromised intracerebral compliance; in such patients the transient decrease in the mean arterial pressure has occasionally been accompanied by a short-lasting reduction of the cerebral perfusion pressure.
Patients on chronic opioid therapy or with a history of opioid abuse may require higher doses.
It is recommended to reduce the dosage in the elderly and in debilitated patients. Opioids should be titrated with caution in patients with any of the following conditions: uncontrolled hypothyroidism, pulmonary disease, decreased respiratory reserve, alcoholism, or impaired hepatic or renal function. Such patients also require prolonged post-operative monitoring. Patients with renal insufficiency should be carefully checked on the symptoms of fentanyl toxicity. As a result of dialysis, the volume of distribution of fentanyl may be altered, which can influence the serum concentrations.

Interaction with neuroleptics
If fentanyl is administered with a neuroleptic, the user should be familiar with the special properties of each drug, particularly the difference in duration of action. When such a combination is used, there is a higher incidence of hypotension. Neuroleptics can induce extrapyramidal symptoms that can be controlled with anti-Parkinson agents. As with other opioids, due to the anticholinergic effects, administration of fentanyl may lead to increases of bile duct pressure and, in isolated cases, spasms of the sphincter of Oddi might be observed. As with all other opioids, fentanyl can have an inhibitory effect on intestinal motility. This should be considered in the pain management of intensive care patients with inflammatory or obstructive intestinal diseases.
In patients with myasthenia gravis, careful consideration should be applied in the use of certain anticholinergic agents and neuromuscular-blocking pharmaceutical agents prior to, and during, the administration of a general anesthetics regimen which includes administering intravenous fentanyl.

Neonatal withdrawal syndrome
If women take opioids chronically during pregnancy, there is a risk that their newborn infants will experience neonatal withdrawal syndrome.

Opioid induced hyperalgesia
Opioid induced hyperalgesia (OIH) is a paradoxical response to an opioid, particularly at high doses or with chronic use, in which there is an increase in pain perception despite stable or increased opioid exposure. It differs from tolerance, in which higher opioid doses are required to achieve the same analgesic effect or treat recurring pain. OIH may manifest as increased levels of pain, more generalized pain (i.e., less focal), or pain from ordinary (i.e. non-painful) stimuli (allodynia) with no evidence of disease progression. When OIH is suspected, the dose of opioids should be reduced or tapered off, if possible.

Risks from Concomitant Use with Benzodiazepines
Profound sedation, respiratory depression, coma, and death may result from the concomitant use of fentanyl injection with benzodiazepines. Observational studies have demonstrated that concomitant use of opioids and benzodiazepines increases the risk of drug-related mortality compared to use of opioids alone. Because of these risks, reserve concomitant prescribing of these drugs for use in patients for whom alternative treatment options are inadequate.

If the decision is made to newly prescribe a benzodiazepine and an opioid together, prescribe the lowest effective dosages and minimum durations of concomitant use.

If the decision is made to prescribe a benzodiazepine in a patient already receiving an opioid, prescribe a lower initial dose of the benzodiazepine than indicated in the absence of an opioid, and titrate based on clinical response.

If the decision is made to prescribe an opioid in a patient already taking a benzodiazepine, prescribe a lower initial dose of the opioid, and titrate based on clinical response.

Follow patients closely for signs and symptoms of respiratory depression and sedation. Advise both patients and caregivers about the risks of respiratory depression and sedation when fentanyl injection is used with benzodiazepines. Advise patients not to drive or operate heavy machinery until the effects of concomitant use of the benzodiaz-epine have been determined. Screen patients for risk of substance use disorders, including opioid abuse and misuse, and warn them of the risk for overdose and death associated with the use of benzodiazepines (See Drug Interactions).

Serotonin Syndrome with Concomitant Use of Serotonergic Drugs
Cases of serotonin syndrome, a potentially life-threatening condition, have been reported during concurrent use of fentanyl injection with serotonergic drugs (See Interactions with Other Medicaments). This may occur within the recommended dosage range.
Serotonin syndrome symptoms may include mental-status changes (e.g. agitation, hallucinations, coma), autonomic instability (e.g. tachycardia, labile blood pressure, hyperthermia), neuromuscular aberrations (e.g. hyperreflexia, incoordination) and/or gastrointestinal symptoms (e.g. nausea, vomiting, diarrhoea) and can be fatal (See Interactions with Other Medicaments). The onset of symptoms generally occurs within several hours to a few days of concomitant use, but may occur later than that. Discontinue fentanyl injection if serotonin syndrome is suspected.

Adrenal Insufficiency
Cases of adrenal insufficiency have been reported with opioid use, more often following greater than one month of use. Presentation of adrenal insufficiency may include non-specific symptoms and signs including nausea, vomiting, decreased appetite, fatigue, weakness, dizziness, and low blood pressure. If adrenal insufficiency is suspected, confirm the diagnosis with diagnostic testing as soon as possible. If adrenal insufficiency is diagnosed, treat with physiologic replacement dosing of corticosteroids. Wean the patient off of the opioid to allow adrenal function to recover and continue corticosteroid treatment until adrenal function recovers. Other opioids may be tried as some cases reported use of a different opioid without recurrence of adrenal insufficiency. The information available does not identify any particular opioids as being more likely to be associated with adrenal insufficiency.

Sexual Function/Reproduction
Long term use of opioids may be associated with decreased sex hormone levels and symptoms such as low libido, erectile dysfunction, or infertility (See Postmarketing Experience/Post marketing Data).

INTERACTIONS WITH OTHER MEDICAMENTS
Central Nervous System (CNS) depressants
Drugs such as barbiturates, benzodiazepines, neuroleptics, general anaesthetics and other, non-selective CNS depressants (e.g. alcohol) may potentiate the respiratory depression of opioids. When patients have received such drugs, the dose of fentanyl required may be less than usual. Concomitant use with fentanyl in spontaneously breathing patients may increase the risk of respiratory depression, profound sedation, coma, and death.

Cytochrome P450 3A4(CYP3A4) inhibitors
Fentanyl, a high clearance drug, is rapidly and extensively metabolized mainly by CYP3A4. When fentanyl is used, the concomitant use of a CYP3A4 inhibitor may result in a decrease in fentanyl clearance. With single-dose fentanyl administration, the period of risk for respiratory depression may be prolonged, which may require special patient care and longer observation. With multiple-dose fentanyl administration, the risk for acute and/or delayed respiratory depression may be increased, and a dose reduction of fentanyl may be required to avoid accumulation of fentanyl. Oral ritonavir (a potent CYP3A4 inhibitor) reduced the clearance of a single intravenous fentanyl dose by two thirds, although peak plasma concentrations of fentanyl were not affected. However, itraconazole (another potent CYP3A4 inhibitor) at 200 mg/day given orally for 4 days had no significant effect on the pharmacokinetics of a single intravenous fentanyl dose. Co-administration of other potent or less potent CYP3A4 inhibitors, such as voriconazole or fluconazole, and fentanyl may also result in an increased and/or prolonged exposure to fentanyl.

Monoamine oxidase inhibitors (MAOIs)
It is usually recommended to discontinue MAOIs 2 weeks prior to any surgical or anaesthetic procedure. However, several reports describe the uneventful use of fentanyl during surgical or anaesthetic procedures in patients on MAOIs.

Muscle relaxants
When fentanyl is used in combination with non-vagolytic muscle relaxants, bradycardia and possibly asystole may occur.

Droperidol
Concomitant use of fentanyl and droperidol can result in higher incidence of hypotension.

Cimetidine
Pre-treatment with, or concurrent administration of, cimetidine may increase plasma levels of fentanyl, when repeated doses of both drugs are used.

Bradycardia may be intensified by pre-treatment with, or concurrent use of, drugs such as beta-blockers, suxamethonium, halothane, vecuronium, which may themselves cause bradycardia.

Effect of fentanyl on other drugs
Following the administration of fentanyl, the dose of other CNS-depressant drugs should be reduced. This is particularly important after surgery, because profound analgesia is accompanied by marked respiratory depression, which can persist or recur in the postoperative period. Administration of a CNS depressant, such as a benzodiazepine, during this period may disproportionally increase the risk for respiratory depression.
The total plasma clearance and volume of distribution of etomidate is decreased by a factor of 2 to 3 without a change in half-life when administered with fentanyl. Simultaneous administration of fentanyl and intravenous midazolam results in an increase in the terminal plasma half-life and a reduction in the plasma clearance of midazolam. When these drugs are co- administered with fentanyl their dose may need to be reduced.

Benzodiazepines
Due to additive pharmacologic effect, the concomitant use of opioids with benzodiazepines increases the risk of respiratory depression, profound sedation, coma and death.

The concomitant use of opioids and benzodiazepines

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increases the risk of respiratory depression because of actions at different receptor sites in the central nervous system that control respiration. Opioids interact primarily at μ-receptors, and benzodiazepines interact at GABAA sites. When opioids and benzodiazepines are combined, the potential for benzodiazepines to significantly worsen opioid-related respiratory depression exists.

Reserve concomitant prescribing of these drugs for use in patients for whom alternative treatment options are inadequate (see Warnings and Precautions).
Limit dosage and duration of concomitant use of benzodiazepines and opioids, and follow patients closely for respiratory depression and sedation.

Serotonergic Drugs
The concomitant use of opioids with other drugs that affect the serotonergic neurotransmitter system has resulted in serotonin syndrome. If concomitant use is warranted, carefully observe the patient, particularly during treatment initiation and dose adjustment. Discontinue fentanyl injection if serotonin syndrome is suspected. Examples of serotonergic drugs are selective serotonin reuptake inhibitors (SSRls), serotonin and norepinephrine reuptake inhibitors (SNRls), tricyclic antidepressants (TCAs), triptans, 5-HT3 receptor antagonists, drugs that affect the serotonin neurotransmitter system (e.g. mirtazapine, trazodone, tramadol), monoamine oxidase (MAO) inhibitors (those intended to treat psychiatric disorders and also others, such as linezolid and intravenous methylene blue) (See Warnings and Precautions).

Incompatibilities
The compatible infusion solutions for intravenous use are as follows:
- Sodium Chloride 0.9% solution
- Glucose 5% solution

This medicinal product must not be diluted with other solutions for parenteral use other than mentioned above.
Compatibility must be checked before administration, if intended to be mixed with other drugs.
Fentanyl citrate is reported to be physically incompatible with pentobarbital sodium, methohexital sodium, thiopental sodium and nafcilline.

PREGNANCY AND LACTATION

Pregnancy
There are no adequate data from the use of fentanyl in pregnant women. Fentanyl can cross the placenta in early pregnancy. Published animal data have shown some reproductive toxicity. The potential risk for humans is unknown.

Chronic use of an opioid during pregnancy may cause drug dependence in the neonate, leading to neonatal withdrawal syndrome.

Administration during childbirth (including caesarean section) is not recommended because fentanyl crosses the placenta and may suppress spontaneous respiration in the newborn period. If fentanyl is administered, assisted ventilation equipment must be immediately available for the mother and infant if required. An opioid antagonist for the child must always be available.

Lactation
Fentanyl is excreted into human milk. Therefore, breast-feeding or use of expressed breast milk is not recommended for 24 hours following the administration of this drug. The risk/benefit of breastfeeding following fentanyl administration should be considered.

ADVERSE EFFECTS

System/Organ Class	Adverse Reaction
Nervous System Disorders	Sedation, Dizziness, Dyskinesia, Headache, Convulsions, Loss of consciousness, Myoclonus, Hyperalgesia
Eye Disorders	Visual disturbance
Cardiac Disorders	Bradycardia, Tachycardia, Arrhythmia, Cardiac arrest
Vascular Disorders	Hypotension, Hypertension, Vein pain, Blood pressure fluctuation, Phlebitis
Respiratory, Thoracic and Mediastinal Disorder	Apnea, Bronchospasm, Laryngospasm, Hiccups, Hyperventilation, Respiratory depression, Cough
Gastrointestinal Disorder	Nausea, Vomiting, Constipation
Skin and Subcutaneous Tissue Disorders	Dermatitis allergic, Pruritis
Musculoskeletal and Connective Tissue Disorders	Muscle rigidity (which may also involve the thoracic muscles)
Injury, Poisoning and Procedural Complications	Confusion postoperative, Anesthetic complication neurological, Agitation postoperative, Airway complication of anesthesia
Psychiatric Disorders	Euphoric mood, Delirium
Immune System Disorders	Hypersensitivity (such as anaphylactic shock, anaphylactic reaction, urticaria)
General Disorders and Administration Site Conditions	Chills, Hypothermia, Drug withdrawal syndrome

When a neuroleptic is used with fentanyl, the following adverse reactions may be observed: chills and/or shivering; restlessness, post-operative hallucinatory episodes; and extrapyramidal symptoms.

Post marketing Data
Serotonin syndrome (See Warnings and Precautions)
Adrenal insufficiency (See Warnings and Precautions)

Androgen deficiency: Cases of androgen deficiency have occurred with chronic use of opioids. Chronic use of opioids may influence the hypothalamic-pituitary-gonadal axis, leading to androgen deficiency that may manifest as low libido, impotence, erectile dysfunction, amenorrhea, or infertility. The causal role of opioids in the clinical syndrome of hypogonadism is unknown because the various medical, physical, lifestyle, and psychological stressors that may influence gonadal hormone levels have not been adequately controlled for in studies conducted to date. Patients presenting with symptoms of androgen deficiency should undergo laboratory evaluation.

Infertility: Chronic use of opioids may cause reduced fertility in females and males of reproductive potential. It is not known whether these effects on fertility are reversible.

SYMPTOMS AND TREATMENT OF OVERDOSE

Symptoms
Overdosage of fentanyl expresses itself by prolongation of the duration of the pharmacological effects. Dependent on the individual sensitiveness, the clinical picture is respiratory depression ranging from bradypnoea to apnoea, bradycardia up to asystole, decrease in blood pressure, circulatory failure, coma, seizure-like activity, muscle rigidity of the chest wall, trunk and extremities, and pulmonary oedema.

Treatment of overdose
Hypoventilation should be treated by administration of oxygen and the patient should be ventilated. Respiratory depression should be treated by administration of an opioid antagonist like naloxone. The usual initial naloxone dose amounts 0.4 to 2 mg. If no effect can be seen, this dose may be repeated every 2 to 3 minutes up to reversal of respiratory depression or awakening. Since the respiratory depressant effect of fentanyl may last longer than the antagonistic effect, repeated doses of naloxone may be appropriate.
Ventilatory problems caused by muscle rigidity can be diminished or abolished by application of a peripherally acting muscle relaxant. The patient should be monitored carefully. Normal body temperature and balanced fluid volumes should be ensured.
In the case of severe and persistent hypotension, hypovolaemia should be considered, which can be compensated by parenteral fluid therapy.

EFFECTS ON ABILITY TO DRIVE AND USE MACHINES

The use of fentanyl may cause a decreased level of reactiveness and concentration. Patients should avoid performance of skilled tasks such as driving or operating machinery for a considerable time (at least 24 hours) after administration of fentanyl. Patients should be accompanied on their way home after discharge and instructed to avoid alcohol.

STORAGE CONDITIONS

Store below 30 C. Protect from light. Retain in carton until time of use.

Following dilution in the recommended diluents, chemical and physical in-use stability has been demonstrated for 24 hours at temperature below 30°C and at refrigerated condition (2°C to 8°C). However, from a microbiological point of view, the product should be used immediately.

SHELF LIFE

3 years.

Following dilution in the recommended diluents, chemical and physical in-use stability has been demonstrated for 24 hours at temperature below 30°C and at refrigerated condition (2°C to 8°C). However, from a microbiological point of view, the product should be used immediately.

DOSAGE FORMS AND PACKAGING AVAILABLE

• 10 x 2mL ampoule (clear glass ampoule)

PRODUCT REGISTRATION HOLDER /MANUFACTURER

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PM 2002068-V

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